Modular Sentence Encoders: Separating Language Specialization from Cross-Lingual Alignment

Yongxin Huang[1], **Kexin Wang**[1], **Goran Glavaš**[2], **Iryna Gurevych**[1]
[1]Ubiquitous Knowledge Processing Lab (UKP Lab)
Department of Computer Science
Technical University of Darmstadt and
National Research Center for Applied Cybersecurity ATHENE, Germany
[2]Center for AI and Data Science, University of Würzburg
[1]www.ukp.tu-darmstadt.de

Abstract

Multilingual sentence encoders (MSEs) are commonly obtained by training multilingual language models to map sentences from different languages into a shared semantic space. As such, they are subject to *curse of multilinguality*, a loss of monolingual representational accuracy due to parameter sharing. Another limitation of MSEs is the trade-off between different task performance: cross-lingual alignment training distorts the optimal monolingual structure of semantic spaces of individual languages, harming the utility of sentence embeddings in monolingual tasks; cross-lingual tasks, such as cross-lingual semantic similarity and zero-shot transfer for sentence classification, may also require conflicting cross-lingual alignment strategies. In this work, we address both issues by means of modular training of sentence encoders. We first train language-specific monolingual modules to mitigate negative interference between languages (i.e., the curse). We then align all non-English sentence embeddings to the English by training cross-lingual alignment adapters, preventing interference with monolingual specialization from the first step. We train the cross-lingual adapters with two different types of data to resolve the conflicting requirements of different cross-lingual tasks. Monolingual and cross-lingual results on semantic text similarity and relatedness, bitext mining and sentence classification show that our modular solution achieves better and more balanced performance across all the tasks compared to full-parameter training of monolithic multilingual sentence encoders, especially benefiting low-resource languages.[1]

1 Introduction

Multilingual Sentence Encoders (MSEs; Artetxe and Schwenk, 2019b; Yang et al., 2020; Reimers and Gurevych, 2020; Feng et al., 2022; Duquenne et al., 2023) embed sentences from different languages into a shared semantic vector space, making them essential tools for multilingual and cross-lingual semantic retrieval (e.g., bitext mining; Schwenk et al., 2021), clustering (e.g., for extractive summarization; Bouscarrat et al., 2019), and filtering (e.g., in content-based recommendation; Hassan et al., 2019), as well as for cross-lingual transfer in supervised text classification (Artetxe and Schwenk, 2019b; Licht, 2023). In this work, we aim to address two limitations in the MSEs through modular training: the curse of multilinguality and the trade-off in performance between different monolingual and cross-lingual tasks.

Like general-purpose multilingual encoder language models (mELMs, e.g., mBERT; Devlin et al., 2019; XLM-R; Conneau et al., 2020), multilingual models specialized for sentence encoding[2] are also subject to the *curse of multilinguality* (CoM; Conneau et al., 2020), a loss of representational precision for each individual language due to sharing of model parameters between many languages, resulting in negative interference (Wang et al., 2020). Training language-specific modules like embedding layers and language adapters (Pfeiffer et al., 2021, 2022) or full models (Blevins et al., 2024) has been proven effective against this issue for general-purpose models, but rarely applied for MSEs, whose sentence embeddings from different monolingual modules need to be semantically aligned to each other. To the best of our knowledge, the only work that targets CoM for MSEs is LASER3 (Heffernan et al., 2022): they train a set of monolingual sentence encoders from scratch through the distillation from a fixed teacher MSE, which is already affected by the CoM.

Existing MSE work mostly focuses on cross-lingual training and evaluation, paying less atten-

[1]Our code is available at https://github.com/UKPLab/acl2025-modular-sentence-encoders.

[2]In fact, many MSEs are derived from mELMs (Reimers and Gurevych, 2020; Feng et al., 2022, *inter alia*) by doing sentence-level training on top of them.

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tion to the monolingual (i.e., within-language) performance, which can be negatively affected by the cross-lingual alignment (Roy et al., 2020). Earlier work on inducing cross-lingual word embeddings (Søgaard et al., 2018; Patra et al., 2019; Glavaš and Vulić, 2020) hints at an explanation for this trade-off: forcing cross-lingual alignment between non-isomorphic monolingual spaces distorts those spaces and thus degrades their monolingual semantic quality. What is more, there also seems to be a trade-off between different cross-lingual tasks: different cross-lingual training approaches yield optimal performance for different tasks. Concretely, MSEs trained on *parallel* data to produce highly similar embeddings for exact translation pairs are effective in bitext mining (Artetxe and Schwenk, 2019b; Feng et al., 2022; Heffernan et al., 2022); however, they perform worse on cross-lingual semantic similarity, failing to produce high similarity for sentences with *similar* but non-equivalent meaning (Reimers and Gurevych, 2020). Conversely, MSEs trained on *paraphrase*[3] data (Yang et al., 2020; Reimers and Gurevych, 2020), i.e. pairs of semantically similar but non-equivalent sentences, yield better semantic similarity performance but are not effective in bitext mining. Paraphrase-trained models also seem to offer weaker performance in zero-shot cross-lingual transfer for sentence classification tasks (Roy et al., 2020), which also seems to benefit more from parallel alignment.

Contributions. In this work, we propose to alleviate all of the above shortcomings by means of *modularity*, that is, parameter separation. As illustrated in Figure 1, we first mitigate the curse of multilinguality by specializing an MSE for each target language, i.e., training language-specific embedding layers and language adapters via masked language modeling (MLM-ing). To obtain high-quality monolingual sentence embeddings, we then train a monolingual sentence encoding adapter (SE adapter) for each language on top of the language adapter, resorting to sentence-level contrastive learning on synthetic monolingual paraphrase data, machine-translated from English. In the next step, we carry out cross-lingual alignment training also in a modular fashion, without jeopardizing the monolingual sentence representation quality. To meet the requirements of different cross-lingual tasks, we train a cross-lingual alignment adapter (CLA adapter) for each non-English language with both cross-lingual *paraphrase* and *parallel* pairs, aligning them to a shared semantic space using English as the pivot language. At inference time, we activate the language-specific modules (embeddings, language adapter, SE adapter, CLA adapter) of the respective language of the input sentence.

Our experiments—encompassing four tasks and 23 linguistically diverse languages and two state-of-the-art MSE models—render our modular approach effective in overcoming the performance trade-offs between both (1) monolingual and cross-lingual tasks as well as (2) different sentence-level tasks types (semantic textual similarity and relatedness on the one side vs. bitext mining and sentence classification on the other), with substantial performance gains over full-parameter training of a single monolithic MSE. Our approach particularly benefits low-resource languages, most affected by the curse of multilinguality. Since both contrastive learning steps in our approach—for monolingual specialization and for cross-lingual alignment—are carried out on machine-translated data, our work also validates the viability of MT for scaling up MSE training data.

2 Related Work

2.1 Multilingual Sentence Embeddings

Multilingual sentence encoders should produce similar sentence embeddings for sentences with similar meaning, regardless whether they come from the same or different languages. Cross-lingual alignment is thus at the core of MSE training, typically achieved by training on parallel data (Artetxe and Schwenk, 2019b; Feng et al., 2022; Duquenne et al., 2023; Gao et al., 2023; Zhao et al., 2024). As a standard practice to acquire high-quality English sentence embedding (Reimers and Gurevych, 2019; Gao et al., 2021), contrastive learning with paraphrase pairs has also been applied to train MSEs. This can be done through teacher-student distillation with an English teacher model trained on English paraphrases (Reimers and Gurevych, 2020; Ham and Kim, 2021), or directly with cross-lingual paraphrases (Wang et al., 2022). Another line of work removes language-specific information to get language-agnostic meaning representation (Yang et al., 2021; Tiyajamorn et al., 2021; Kuroda et al., 2022). To the best of our knowledge, our work is the first attempt to address multiple conflicting

[3]We use the word "paraphrase" in a broad sense, to include also, e.g., entailment pairs or question-answer pairs.

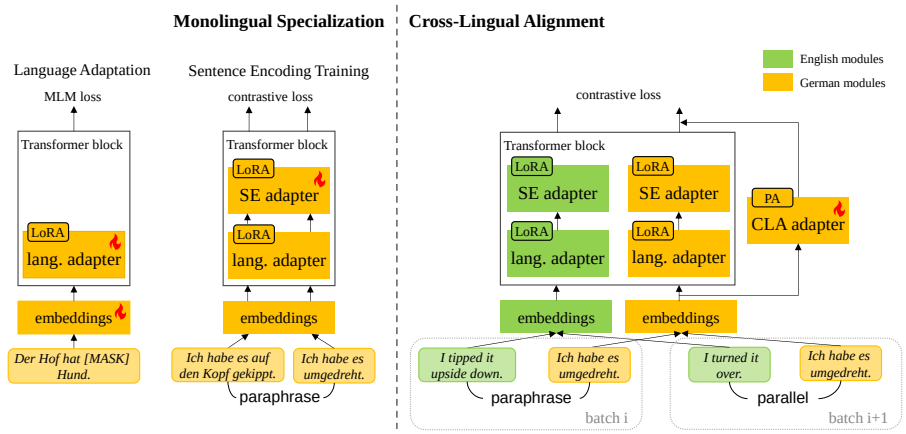


Figure 1: Illustration of how we apply our modular training to a pre-trained multilingual sentence encoder. In each step, only the module marked with the fire symbol is trained. In the monolingual specialization step, we train a language-specific embedding layer, a language adapter and a monolingual sentence encoding (SE) adapter for each language. In the cross-lingual alignment (CLA) step, the monolingual (e.g., German) representation is aligned to the English representation via cross-lingual paraphrase and parallel data, in alternate batches. PA: parallel adapter.

factors in MSE training, aiming to yield optimal performance trade-off across a variety of tasks.

2.2 Lifting the Curse of Multilinguality

The post hoc parameter-efficient adaptation for individual languages is mostly done for on feneral-purpose mELMs like mBERT and XLM-R (Pfeiffer et al., 2020, 2021; Parović et al., 2022, *inter alia*) through continued pre-training on the target language corpora. Expanding or replacing multilingual vocabulary with target language tokens and smart initialization of their embeddings (Chau and Smith, 2021; Pfeiffer et al., 2021; Minixhofer et al., 2022; Dobler and de Melo, 2023) has been shown to improve sample efficiency of post hoc language adaptation of multilingual models.

However, previous adapter-based methods for general-purpose models do not address the unique challenges posed by MSEs, as MSE training additionally requires specialization for sentence encoding after the standard pre-training. Besides, existing methods are only used in tasks where the input is always in one language, so only one specific language adapter needs to be activated (Pfeiffer et al., 2020). In contrast, MSEs often deal with cross-lingual sentence pairs as input (in cross-lingual STS or bitext mining), which requires explicit alignment training of language-specific adapters. In the field of MSEs, the language-specific adaptation still relies on monolithic full-parameter training of the whole model: either trained only for a certain language (Mohr et al., 2024), or distilled from a massively multilingual teacher model which is already affected by the curse of multilinguality and never really trained to model fine-grained semantic similarity (Heffernan et al., 2022). Some existing MSE efforts (Mao et al., 2021; Kuroda et al., 2022; Liu et al., 2023; Yano et al., 2024) do leverage lightweight modules for cross-lingual training, but these modules are still (massively) multilingual, i.e., do not alleviate the curse of multilinguality.

3 Modular Sentence Encoder

Our main objective is to obtain multilingual sentence embeddings that excel across the board, despite the conflicts between different tasks and scenarios: (i) in both monolingual and cross-lingual tasks, despite cross-lingual semantic alignment possibly being at odds with monolingual semantic specialization; and (ii) in different types of cross-lingual tasks, despite the fact that they require different types of cross-lingual alignment training (Roy et al., 2020). To mitigate these inherent trade-offs, we propose a modular approach, i.e., to isolate parameters for different requirements, as illustrated in Figure 1: we train a set of language-specific modules to (i) specialize the MSE for each individual language, and (ii) to align the monolingually adapted MSEs for cross-lingual tasks.

3.1 Monolingual Specialization

We specialize MSEs like LaBSE (Feng et al., 2022) and multilingual E5 (Wang et al., 2024) for each language by training language-specific (i) embedding layers and (ii) adapters with monolingual data.

Language Adaptation (LA). For each language, we train a new, language-specific tokenizer and initialize its new embedding matrix following the FOCUS approach (Dobler and de Melo, 2023). In a nutshell, FOCUS copies the embeddings for tokens that already exist in the vocabulary of the original MSE; for new tokens, it interpolates between embeddings of similar tokens from the original vocabulary. Compared to random initialization, FOCUS keeps a substantial amount of information from the pre-trained embeddings of the multilingual model in the new embeddings, making them "compatible" with the model body, avoiding the need to train them from scratch for each language: this leads to more sample efficient training for the embedding layers.[4] For each target language, we then do standard (continued) MLM-ing on the monolingual corpora of the language. To this end, we resort to modular, parameter-efficient fine-tuning (PEFT): besides the parameters of the new embedding matrix, we train only the low-rank adaptation matrices (LoRA; Hu et al., 2022) in encoder's layers. PEFT has been widely adopted for post-hoc language specialization of vanilla mELMs (Pfeiffer et al., 2020, 2021; Parović et al., 2022).

Sentence Encoding (SE) (Re-)Training. As a token-level objective, (continued) MLM-ing is detrimental to the original sentence embedding abilities of a pre-trained MSE: we thus need to re-specialize the model for (monolingual) sentence encoding: for this, we use a standard contrastive learning objective, Multiple Negative Ranking Loss (MNRL; Henderson et al., 2017), and train on the (noisy) monolingual paraphrase data, machine-translated from English. This step is also done in a modular way by stacking another set of monolingual adapters (again LoRA), the *SE adapter*, on top of the LA. In this training step, only the parameters of the SE adapter are updated, in order to obtain the monolingual sentence encoding ability; the encoder body, language-specific embeddings layer and the previously trained LA are all kept frozen.

3.2 Cross-Lingual Alignment (CLA)

The mutually independent language adaptation for individual languages warrants a cross-lingual sentence-level alignment step, so that the sentence embeddings can also be used in cross-lingual applications. To prevent negative interference between cross-lingual alignment and previously imparted monolingual SE abilities, we train a cross-lingual alignment (CLA) module as a *parallel adapter* (He et al., 2022) for each non-English language. Since our machine-translated monolingual paraphrase datasets are parallel across all languages, we can create both cross-lingual *paraphrase* pairs (i.e. sentence in language A and its paraphrase in language B) and *parallel* pairs (i.e. sentence in language A and its direct translation in language B), which can be combined in training to mitigate the inherent interference between semantic similarity, bitext mining and cross-lingual transfer for classification (see §1).

All the cross-lingual training pairs consist of one sentence in English and another sentence in the target language. To align the non-English sentence embeddings to the English ones, we alternate training on a batch of paraphrase data with the same MNRL—just like in monolingual SE training—and another batch of parallel data with the cosine similarity loss (following Heffernan et al., 2022). The cross-lingual alignment training updates only language-specific CLA adapters; the monolingual modules of the corresponding input language are activated in the forward pass, but not updated.

We favor bilingual alignment with English over multilingual alignments[5], because English embeddings are the most reliable: not only is the initial multilingual encoder most "fluent" in English, but we also trained English embeddings on gold paraphrase data, whereas all other SE adapters are trained with noisy translations. Because of this, we omit to train the CLA adapter for English: with English embedding space being of the best semantic quality, we want embeddings from other languages to adapt (through their CLA adapters) to the English space, and not vice versa. Using English as a pivot has already been proven effective in aligning non-English languages to each other (Reimers and Gurevych, 2020; Heffernan et al., 2022). We also do an empirical comparison between bilingual and all-pair alignment in §6.1.

3.3 Inference

After training, we have several modules for each language: embedding layer, language adapter, SE adapter and CLA adapter. When encoding the input text, the corresponding modules for the input language should be activated. Thus, the language

[4]We refer the reader to the original paper for more details.

[5]Given the multi-parallel nature of the paraphrase data we obtained with MT, direct alignment between all non-English language pairs is possible.

of the input text should be known. Otherwise, one can easily apply any SotA language identification models (Kargaran et al., 2023) to detect the input language first.

4 Experimental Setup

4.1 Models

We start from two popular MSEs as base models for our modular specialization: **LaBSE** and multilingual E5-base (**mE5**). LaBSE has been pretrained on billions of parallel sentence pairs (Feng et al., 2022). Starting from XLM-R-base (Conneau et al., 2020), mE5 has first been trained on around 1 billion of (noisy) weak-supervision pairs, then on around 1.6 million high-quality sentence pairs (Wang et al., 2024). The goal of our work is *not* to outperform *other* MSEs or achieve SotA performance; instead, we aim to show that our proposed modular specialization offers clear benefits over monolithic full-model training.

Monolithic Baselines. Our primary baseline is the monolithic MSE model for which all parameters are updated in each training step, akin to mSimCSE (Wang et al., 2022). While mSimCSE originally trains only on (English or cross-lingual) NLI data, we extend this to make the comparison with our modular variants as fair as possible: we use all the MT-obtained multilingual paraphrase datasets (beyond just NLI) as in our modular training. We have the following monolithic-model variants: (i) $\textbf{Full}_{\textbf{en}}$, trained only on (clean) English paraphrase data; (ii) $\textbf{Full}_{\textbf{m}}$, trained only on monolingual data of all languages (each batch is monolingual, language randomly sampled for each batch); (iii) $\textbf{Full}_{\textbf{c}}$, trained only on cross-lingual paraphrase pairs (the language for each sentence in a paraphrase pair is randomly selected); and (iv) $\textbf{Full}_{\textbf{mc}}$, trained sequentially, first on monolingual and then on cross-lingual paraphrases.

Modular Variants. We evaluate the following variants: (i) $\textbf{Mod}_{\textbf{en}}$, as a baseline: a monolingual SE adapter is trained only on English paraphrase data and used for all other languages; i.e., we transfer the sentence encoding ability from English; (ii) $\textbf{Mod}_{\textbf{m}}$: with only monolingual specialization, i.e. a monolingual SE adapter is trained with paraphrase dataset for every language; (iii) $\textbf{Mod}_{\textbf{mc-pp}}$ adds a CLA adapter trained only on cross-lingual ***paraphrase*** data to Mod_{m}; (iv) $\textbf{Mod}_{\textbf{mc-pl}}$ adds a CLA adapter trained only on cross-lingual ***parallel*** data to Mod_{m}; (v) $\textbf{Mod}_{\textbf{mc-jt}}$ is our complete setting with a CLA adapter trained ***jointly*** on both paraphrase and parallel data. We do the modular training on LaBSE for 23 languages present in the evaluation datasets. Due to the intensive LA step and limited resources, for mE5 we train the modules for a subset of 10 languages.[6]

4.2 Training Data

Supervised paraphrase data is crucial for achieving high performance in sentence embedding tasks, yet a large amount of such data is only available in English. Compared to the labor-intensive manual mining and labelling or translation of paraphrase data in all languages, machine translation is significantly more cost-effective and scalable. The SotA MT models today can already provide high-quality translation for hundreds of languages, including very low-resource ones (NLLB Team et al., 2022; Kudugunta et al., 2023). This motivates us to translate, with NLLB 3.3B as our MT model (NLLB Team et al., 2022), five English paraphrase datasets—MNLI (Williams et al., 2018), Sentence-Compression (Filippova and Altun, 2013), SimpleWiki (Coster and Kauchak, 2011), Altlex (Hidey and McKeown, 2016) and QuoraDuplicateQuestions[7], containing combined around 600K sentence pairs—into all 22 languages found in our downstream evaluation datasets. This results in a multi-parallel paraphrase dataset spanning 23 languages, from which we create instances for monolingual and cross-lingual training.

We train language-specific tokenizers and carry out monolingual language adaptation on monolingual corpora combined from language-specific portions of CC100 (Conneau et al., 2020) and MADLAD-400 (Kudugunta et al., 2023).

4.3 Evaluation Data

We evaluate the obtained sentence encoders on four tasks: semantic textual similarity (STS), semantic textual relatedness (STR), bitext mining, and sentence classification. For the first three tasks, we do evaluation in the "zero-shot" setup, i.e., without any task-specific supervised training. We only evaluate on high-quality datasets, compiled either manually from scratch or by human post-editing of machine translations.[8]

[6] We provide the full list of languages in Appendix A and training details in Appendix B.

[7] See Appendix C.1 for details on the training datasets

[8] See Appendix C.2 for details on the evaluation datasets.

Semantic Textual Similarity. The models need to produce a score indicating semantic similarity for a pair of sentences. We simply use the cosine similarity between the embeddings of the sentences. Performance is reported as Spearman correlation ($\times 100$) against human scores. We collect existing multilingual STS datasets and use parallel monolingual STS data to create high-quality cross-lingual evaluation pairs. For example, the STS datasets for Czech, German and French (Hercig and Kral, 2021) and the datasets for Dutch, Italian and Spanish (Reimers and Gurevych, 2020) are parallel to each other, as they are translated from the same **STS17** (Cer et al., 2017) English data. The same applies for the STS datasets for Turkic languages in Kardeş-NLU (Senel et al., 2024) and the Korean STS dataset from Ham et al. (2020), all translated from the English STS-Benchmark (**STSB**; Cer et al., 2017). We can thus leverage this effectively multi-parallel STS data for cross-lingual evaluation on many more language pairs, including pairs never evaluated in prior work, e.g. Czech-Italian or Korean-Uzbek.

Semantic Textual Relatedness. Semantic relatedness is a broader concept than similarity, that also considers aspects like topic or view similarity (Ousidhoum et al., 2024). We use the same metric as in the STS task. Similar to STS, we aggregate the multi-parallel monolingual data and create cross-lingual pairs between Polish (Dadas et al., 2020), Dutch (Wijnholds and Moortgat, 2021), and Spanish (Araujo et al., 2022), all translated from the English **SICK** dataset (Marelli et al., 2014). **STR24** (Ousidhoum et al., 2024) contains monolingual STR data for low-resource African and Asian languages; but it is not multi-parallel, and as such only lends itself to monolingual evaluation.

Bitext Mining. The model should mine parallel sentences (translation pairs) from two lists of monolingual sentences based on the cosine similarity of bilingual sentence pairs. Following Heffernan et al. (2022), we use the `xsim` score (error rate of wrongly aligned sentences; Artetxe and Schwenk, 2019a) to evaluate our models on two bitext mining datasets: **FLORES** (Goyal et al., 2022) and **Tatoeba** (Artetxe and Schwenk, 2019b). We only evaluate on the languages for which we have trained language-specific modules. Since FLORES is multi-parallel, we test on all possible language pairs between our target languages. Tatoeba only contains English-X data: we average the results from both mining directions (English→X and X→English) for all languages X.

Topic Classification. We resort to **SIB**-200 (Adelani et al., 2024) to obtain data for topical sentence classification for our 23 target languages. In monolingual evaluation, we train a simple Logistic Regression (Cox, 1958) classifier on top of our frozen sentence encoder for each target language. In (zero-shot) cross-lingual transfer setup, we train the classifier only on English data and use it for other languages.

Alignment Metrics. In standard task formulations, cross-lingual STS is *bilingual*, i.e., a sentence in one language is compared only against sentences in one (and same) other language. Such an evaluation setup fails to capture the **language bias** of an MSE (Roy et al., 2020): in a multilingual candidate pool, the model might prefer certain language (pair) over others, e.g., map sentences from the same language closer in the embedding space even if they are semantically dissimilar. Following Reimers and Gurevych (2020), we quantify language bias as the performance drop when switching from bilingual to *multilingual* evaluation, for which we calculate the Spearman correlation on the concatenation of all bilingual datasets. To this end, we use the multi-parallel STSB and SICK datasets; we report the difference between the average performance on all individual bilingual tasks and the performance on the single multilingual task. Another indicator of semantic quality of multilingual representation spaces is the similarity of monolingual semantic structures, i.e., the degree of their isomorphism. It can be quantified by Relational Similarity (**RSIM**; Vulić et al., 2020) on a bilingual parallel corpus: we calculate the corresponding sets of cosine similarity scores for all monolingual sentence pairs in each of the two languages, and report RSIM as Pearson correlation between the two sets of corresponding monolingual cosines. We measure RSIM on FLORES, averaging the results across all language pairs.

5 Results

We report the results for our LaBSE-based models in Table 1 and for mE5-based models in Table 2.

5.1 Full Model Results

Further training on monolingual paraphrase data (Full_{en} and Full_{m}) can already largely improve the original models' (first row in each table) perfor-

	Monolingual tasks					Cross-lingual tasks						Alignment metrics		
	STS↑		**STR↑**		**CLS↑**	**STS↑**		**STR↑**	**CLS↑**	**Bitext Mining↓**		**Language Bias↓**		**RSIM↑**
Dataset	sts17	stsb	sick	str24	sib	sts17	stsb	sick	sib	flores	tatoeba	stsb	sick	flores
LaBSE	76.7	71.9	68.0	69.2	82.7	74.5	64.4	63.8	83.6	0.14	3.87	1.02	2.32	0.64
Full_{en}	82.7	**80.9**	**76.5**	75.4	84.1	78.8	71.5	70.4	83.5	0.49	4.72	0.87	1.48	0.70
Full_{m}	**82.9**	80.4	76.4	**75.9**	84.8	**79.4**	71.5	70.9	83.9	0.29	4.43	0.88	1.27	0.74
Full_{c}	81.0	79.1	75.1	75.3	85.1	77.8	72.1	71.5	85.3	**0.20**	**4.00**	**0.53**	0.70	**0.77**
Full_{mc}	80.0	79.2	75.1	75.4	**86.0**	76.7	**72.7**	**71.7**	**86.3**	0.21	4.17	**0.53**	**0.64**	**0.77**
Mod_{en}	82.6	**82.1**	76.3	**78.7**	84.9	80.1	74.8	71.5	83.6	0.16	3.68	0.90	1.24	0.73
Mod_{m}	**83.1**	**82.1**	76.5	78.4	85.5	80.6	75.3	71.9	85.0	**0.15**	3.63	1.05	1.16	0.75
$\text{Mod}_{\text{mc-pp}}$	82.9	81.8	**76.7**	77.5	**86.0**	**80.7**	76.0	**72.8**	85.0	0.16	**3.49**	0.71	0.92	0.76
$\text{Mod}_{\text{mc-pl}}$	81.4	81.6	76.0	77.2	85.8	79.1	76.1	72.4	**86.2**	**0.15**	3.64	**0.56**	**0.67**	**0.82**
$\text{Mod}_{\text{mc-jt}}$	82.7	**82.1**	76.6	78.1	85.8	80.3	**76.4**	72.7	85.7	**0.15**	3.55	**0.56**	0.78	0.79
Ablations														
Mod_{m} w/o LA	81.3	78.1	74.3	75.9	84.0	79.0	72.0	71.0	84.7	0.13	3.84	0.85	1.10	0.75
$\text{Mod}_{\text{c-jt}}$	82.7	81.9	76.4	77.6	85.3	80.3	76.0	72.6	85.3	0.16	3.69	0.58	0.88	0.79

Table 1: Results of the LaBSE-based models for 23 languages. Reported results are averages over all languages in each evaluation dataset. The best result within the Full group and the Mod group on each dataset is denoted in **bold**. The second-best result in the Mod group is underlined. CLS stands for classification. See detailed results on each individual language (pair) in Appendix D.

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mance on all tasks, except for bitext mining. The off-the-shelf LaBSE model is a strong baseline for bitext mining, as it has been pre-trained on a massive amount of parallel data, which perfectly aligns with the goal of bitext mining. This confirms the previous finding that training on paraphrase data can disturb bitext mining ability (Reimers and Gurevych, 2020). Full_{m} trained on MT-ed monolingual data in all target languages outperforms Full_{en} (i.e., the $\text{mSimCSE}_{\text{en}}$ setting in Wang et al., 2022) slightly on LaBSE and significantly on mE5, demonstrating the limitation of cross-lingual transfer of sentence-embedding specialization from English, especially if the base model has not been subjected to massive cross-lingual pre-training on parallel data like LaBSE. The improved results of Full_{m} also indicate that machine translation is a reliable alternative to the labor-intensive labelling of training data for a broad range of languages.

Full_{m} outperforms Full_{c} on monolingual STS and STR tasks, whereas the opposite is true in cross-lingual tasks: this confirms the inherent trade-off between monolingual and cross-lingual abilities of MSEs. The inability of monolingual training, even using multi-parallel data, to induce strongly aligned cross-lingual semantic structures is confirmed by the higher language bias and lower RSIM scores of Full_{m}. The trade-off between monolingual and cross-lingual performance is more pronounced in mE5 results. The sequential combination of both monolingual and cross-lingual training (Full_{mc}) is unable to resolve the conflict and yields results similar to Full_{c}: in a monolithic MSE model, the subsequent cross-lingual alignment seems to distort the semantic quality of monolingual subspaces. One notable exception is *monolingual* text classification, where Full_{mc} outperforms Full_{m} on LaBSE. We speculate that is because topic classification relies on lexical cues rather than fine-grained sentence meaning: cross-lingual training probably improves lexical alignments and the fine-grained distortions it brings to monolingual semantics play no role in this semantically coarse task.

5.2 Modular Model Results

Monolingual Training. We first compare the baseline Mod_{en}, with an SE adapter trained only on English data and shared across all languages, against Mod_{m}, with a language-specific SE adapter for each language. As is the case for monolithic models, Mod_{m} with language-specific sentence encoding training with noisy machine-translated data outperforms the transfer from English-only SE training (Mod_{en}) on mE5's cross-lingual tasks, dramatically reducing the language bias. Looking at performance on monolingual tasks, our Mod_{m} with monolingual specialization (LA and SE) successfully mitigates the curse of multilinguality, which seems to be present in its monolithic counterpart Full_{m}: the gains are particularly prominent on monolingual STSB (+1.7 on LaBSE, +2.5 on mE5) and STR24 (+2.5 on LaBSE), datasets that encompass most low-resource languages. The importance of modularity becomes most apparent on *cross-lingual* STS and STR, where our Mod_{m}, not exposed to any explicit cross-lingual alignment, outperforms the explicitly cross-lingually trained

	Monolingual tasks			Cross-lingual tasks					Alignement metrics		
	STS↑	**STR↑**	**CLS↑**	**STS↑**	**STR↑**	**CLS↑**	**Bitext Mining↓**		**Language Bias↓**		**RSIM↑**
Dataset	stsb	sick	sib	stsb	sick	sib	flores	tatoeba	stsb	sick	flores
mE5	72.5	74.2	74.0	54.1	61.0	73.5	1.85	9.89	23.22	12.11	0.60
Full_{en}	75.8	75.4	83.4	55.4	62.2	82.9	1.46	9.98	7.21	5.79	0.59
Full_{m}	**79.6**	**75.5**	85.5	60.2	64.1	85.2	0.62	7.85	2.60	3.16	0.67
Full_{c}	77.7	73.9	**85.6**	**66.7**	**67.7**	85.5	**0.26**	6.37	1.11	1.24	**0.74**
Full_{mc}	77.4	73.1	85.4	**66.7**	66.9	**86.5**	**0.26**	**6.33**	**1.05**	**1.14**	**0.74**
Mod_{en}	79.9	75.8	87.0	66.2	66.7	87.0	0.26	5.81	6.66	5.27	0.72
Mod_{m}	**82.1**	75.4	87.8	69.8	68.5	87.7	0.22	5.27	2.82	3.07	0.74
$\text{Mod}_{\text{mc-pp}}$	81.7	**76.4**	87.9	73.2	70.5	87.6	0.20	5.19	1.58	2.08	0.75
$\text{Mod}_{\text{mc-pl}}$	80.8	75.2	**88.5**	72.8	69.6	**89.0**	0.22	5.61	2.15	2.05	**0.82**
$\text{Mod}_{\text{mc-jt}}$	81.9	**76.4**	88.3	**73.8**	**70.7**	88.3	**0.19**	**5.00**	**1.33**	**1.73**	0.80
Ablations											
Mod_{m} w/o LA	80.8	76.0	87.2	61.5	64.4	86.3	0.56	7.63	3.87	3.53	0.68
$\text{Mod}_{\text{c-jt}}$	81.9	76.2	87.7	73.9	70.6	88.2	0.19	5.24	1.39	1.93	0.80

Table 2: Results of the mE5-based models for 10 languages. Reported results are averages over all languages in each evaluation dataset. The best result within the Full group and the Mod group on each dataset is denoted in **bold**. The second-best result in the Mod group is underlined. CLS stands for classification. See detailed results on each individual language (pair) in Appendix D.

monolithic variants (Full_{c} and Full_{mc}). This shows that monolingual training on multi-parallel data leads to semantic alignment, emphasizing the potential of MT for synthesizing MSE training data. Our Mod variants also have a clear advantage over monolithic (Full) models in *bitext mining* (both for LaBSE and mE5), even in the absence of explicit cross-lingual training (i.e., Mod_{m}). This suggests that multilingual training on full models messes up not only the monolingual spaces (i.e., the curse of multilinguality) but also the cross-lingual relations, which is alleviated by our modular approach.

Cross-Lingual Training. Adding cross-lingual alignment in a modular fashion (Mod_{mc} variants) brings further gains (compared to Mod_{m}) in cross-lingual tasks. Cross-lingual adapters, either trained on paraphrase data ($\text{Mod}_{\text{mc-pp}}$) or parallel data ($\text{Mod}_{\text{mc-pl}}$) can effectively reduce language bias and increase isomorphism of monolingual spaces (cf. Mod_{m}). Results further show that paraphrase- and parallel-CLA adapters benefit different types of cross-lingual tasks. On both LaBSE and mE5, $\text{Mod}_{\text{mc-pl}}$ has the strongest performance in cross-lingual classification transfer (CLS), which correlates with the degree of isomorphism (RSIM). However, adding this CLA adapter trained with parallel data has a negative impact on the monolingual performance (cf. Mod_{m}). Conversely, $\text{Mod}_{\text{mc-pp}}$ is better at both monolingual and cross-lingual STS/STR than $\text{Mod}_{\text{mc-pl}}$. This confirms the conflicting requirements of different downstream tasks. Combining both training strategies in $\text{Mod}_{\text{mc-jt}}$ mitigates individual shortcomings of $\text{Mod}_{\text{mc-pp}}$ and $\text{Mod}_{\text{mc-pl}}$, resulting in well-balanced performance across all tasks, including the monolingual ones. Our complete $\text{Mod}_{\text{mc-jt}}$ setup thus makes the best use of our multi-parallel paraphrase dataset.

5.3 Ablation of Monolingual Specialization

Additional monolingual training for each language as an intermediate step before cross-lingual alignment distinguishes our modular approach from other popular MSE training strategies. We thus ablate the contribution of the monolingual specialization step (last two rows in Table 1 and Table 2).

Language Adaptation. We first remove the LA step, i.e. we omit the MLM training with language-specific embedding layer and language adapter and directly train the monolingual SE adapter on the original MSE. For both LaBSE and mE5, this leads to a significant performance drop compared with Mod_{m}. Without language adaptation, adapter-based SE training even underperforms Full_{m} in monolingual tasks on LaBSE. But it can still improve over Full_{m} in cross-lingual tasks: this again suggests that modular multi-parallel monolingual SE training benefits cross-lingual semantic alignment more than multilingual training on shared full-model parameters.

Monolingual SE Training. To isolate the contribution of the monolingual SE adapter, we remove the SE adapter for non-English languages from $\text{Mod}_{\text{mc-jt}}$ to get a $\text{Mod}_{\text{c-jt}}$ baseline: now the sentence encoding in other languages is learned only through the alignment to the English representations. We observe a slight drop in both monolingual and cross-lingual tasks and an increase in language bias, suggesting that the removal of monolingual SE training is detrimental to the strong cross-lingual alignment of language-specific representa-

	monoling.	cross-ling.	lang. bias
$\text{Mod}_{\text{mc-pp}}$	81.8	76.0	0.71
$\text{Mod}_{\text{mc-pp}}$ all-pair	80.9	75.7	1.25
$\text{Mod}_{\text{mc-pl}}$	81.6	76.1	0.56
$\text{Mod}_{\text{mc-pl}}$ all-pair	75.8	72.2	1.38

Table 3: STSB results of LaBSE-based Mod variants with our standard English-centric alignment ($\text{Mod}_{\text{mc-pp}}$ and $\text{Mod}_{\text{mc-pl}}$) and the alternative all-pair alignment.

step	module	size	time
language adaptation	embedding layer language adapter	8.15% 0.09%	20h
sentence encoding	SE adapter	0.25%	15m
cross-lingual alignment	CLA adapter	1.50%	30m

Table 4: Size (percentage of the original LaBSE size of 472M parameters) and training time for each module on an A100 40G GPU. See Appendix B for training details.

tion subspaces. The ablation results prove that our monolingual specialization steps are not only effective for improving monolingual performance of individual languages, but also plays an indispensable role in cross-lingual alignment.

6 Discussion

6.1 All-Pair or English-Centric Alignment

We provide an additional experiment to empirically show the advantage of alignment to English representations over all-pair alignment on a subset of 7 languages (the STSB languages). In all-pair alignment, all languages are aligned with each other instead of only to English. For both cross-lingual training on paraphrase data and parallel data, the results in Table 3 show a clear performance drop in both monolingual and cross-lingual evaluation, indicating that due to the increased complexity of aligning every language pair directly, the all-pair alignment without a fixed pivot can reduce the representation quality of each language as well as the alignment between languages.

6.2 Efficiency of our Method

The parameter size of each module and training time for each step are reported in Table 4. One limitation of our method is that the parameter size of language-specific modules scales linearly with the number of languages. However, there is no "free lunch" in addressing the curse of multilinguality. In contrast to training monolingual full models, we try to achieve a balance between performance and model size. Our modular design offers high flexibility, supporting diverse use cases. It is unlikely that all application scenarios involve hundreds of languages simultaneously. For use cases with several or even a single language, only the relevant modules need to be loaded, which minimizes the computational and memory overhead. Additionally, our method allows new languages to be added independently, without the need for retraining the backbone or any of the previously trained modules.

Our modular approach to multilingual sentence encoding presented in this work opens a range of possibilities for further (modular) improvements. Though we reduce the vocabulary size by switching from the original multilingual tokenizers (501K for LaBSE and 250K for mE5) to 1/10 and 1/5 (50K for each of our monolingual tokenizer), our embedding layers remain the largest contributor to the model weights (Table 4). Further parameter reduction can be promising directions for future work. For instance, even smaller vocabulary sizes can be tried out, as some of the LASER3 models use as small as 8K vocabulary (Heffernan et al., 2022). Additionally, LoRA could also be applied to the embedding layer to compress the module. Finally, modules can be trained for language families instead of individual languages, reducing the number of required parameters while leveraging linguistic similarities.

7 Conclusion

Multilingual sentence encoders encode sentences from many languages in a shared semantic space. As a consequence, they suffer from the curse of multilinguality and trade monolingual performance for cross-lingual alignment. Moreover, the choice of different types of training data (paraphrases vs. parallel data) results in performance trade-offs between cross-lingual downstream tasks. In this work, we addressed these shortcomings via modularity. We first specialize a multilingual sentence encoder to individual languages by training language-specific embedding layers, language adapters and monolingual sentence encoding adapters. The high-quality monolingual sentence embedding spaces are then aligned to a shared space through another set of cross-lingual alignment adapters, trained jointly on both paraphrase and parallel data. We show (i) that this modular approach yields gains w.r.t. both monolingual and cross-lingual performance, and (ii) that machine-translated data can help train effective sentence encoders.

Limitations

We only experiment with encoder-based MSEs like LaBSE and mE5. Though this is the mainstream architecture for most MSEs, there are also pre-trained MSEs with the encoder-decoder architecture (Duquenne et al., 2023). Since the pre-training training objectives of such models are different from the encoder-based models we use (i.e. MLM and contrastive sentence embedding learning), our current modular training approach cannot be directly applied to them without adaptations. We thus leave the application of our modular approach to improve encoder-decoder MSEs to future work.

Having language-specific modules for each language requires that the language of the input text is known. If the language is unknown, a prior language identification step is needed to determine it, as we do not have a built-in language detection module. Fortunately, language identification is generally straightforward and reliable models that recognize hundreds of languages are readily available (Kargaran et al., 2023).

Ethics Statement

Our experiments use publicly available datasets and benchmarks for training and evaluation: these are all commonly used in the NLP research. No personal information or sensitive data are involved in our work. Existing biases in the public datasets, our machine-translated datasets and pre-trained models can still be relevant concerns, as we do not specifically mitigate them in this work.

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A Languages

Table 5 lists the languages with their codes and scripts.

Code	Language	Script
am	Amharic	Ge'ez
ar	Arabic	Arabic
az	Azerbaijani	Latin
cs	Czech	Latin
de	German	Latin
en	English	Latin
es	Spanish	Latin
fr	French	Latin
ha	Hausa	Latin
it	Italish	Latin
kk	Kazakh	Cyrillic
ko	Korean	Hangul
ky	Kyrgyz	Cyrillic
mr	Marathi	Devanagari
nl	Dutch	Latin
pl	Polish	Latin
ru	Russian	Cyrillic
rw	Kinyarwanda	Latin
te	Telugu	Ge'ez
tr	Turkish	Latin
ug	Uyghur	Arabic
uz	Uzbek	Latin
zh	Chinese	Han (simplified)

Table 5: Languages with their code used in this paper and the scripts.

B Training Details

The pre-trained models and libraries used in our experiments are listed in Table 6. They are used only for research purposes in this work. We do not do specific hyperparameter tuning because of the large-scale MLM training and the robustness of contrastive learning against hyperparameters (Wang et al., 2022). Thus, we mainly use hyperparameters recommended by the previous work or default settings in the packages.

B.1 Full-Parameter Baselines

Both monolingual and cross-lingual contrastive learning on all baselines are done with a sequence length of 128, batch size of 128 and learning rate of 2e-5. To make a fair comparison with the modular variants, we train Full_{m} and Full_{c} for 3 epochs on the 600K monolingual or cross-lingual paraphrase data, respectively, while the Full_{mc} is obtained by 3 epochs of monolingual training followed by another 3 epochs of cross-lingual training. We found

Model	HuggingFace Name	License
LaBSE	sentence-transformers/LaBSE	apache-2.0
NLLB	facebook/nllb-200-3.3B	cc-by-nc-4.0
mE5 base	intfloat/multilingual-e5-base	mit
Libarary	**GitHub Link**	**License**
transformers	https://github.com/huggingface/transformers	apache-2.0
sentence-transformers	https://github.com/UKPLab/sentence-transformers	apache-2.0
adapters	https://github.com/adapter-hub/adapters	apache-2.0
deepfocus	https://github.com/konstantinjdobler/focus	mit

Table 6: Models and libraries used in our experiments.

that further increasing the number of epochs will not improve the performance.

B.2 Modular Training

FOCUS The training of language-specific tokenizers and the initialization of language-specific embedding matrices is done using the `deepfocus` package (Table 6). We set the vocabulary size to 50K for each language. The dimensionality of fastText embeddings used to calculate token similarity is set to 300 as recommended. Other parameters remain as default. We use up to 10M sentences for the training of the tokenizer and the auxiliary fastText embeddings on each language.

Language Adaptation As the language adapter, we use a LoRA adapter (Hu et al., 2022) on key, query, value matrices of the attention layers, with a rank of 8, alpha of 16 and 0.1 dropout. For each language, we train the embedding layer and the language adapter for 200K steps, with a batch size of 128. The training is done in bf16 precision. For high-resource languages, 200K steps of training only cover a small portion of the available data in MADLAD-400 (Kudugunta et al., 2023). For low-resource languages, we use all data of the corresponding language from CC100 (Conneau et al., 2020) and MADLAD-400 (Kudugunta et al., 2023).

Monolingual Sentence Encoding For the monolingual SE training, we use a LoRA adapter (Hu et al., 2022) on all linear layers, with a rank of 8, alpha of 16 and 0.1 dropout. We use the 600K paraphrase data in the corresponding language for contrastive sentence embedding training for each language, with a sequence length of 128, batch size of 128 and learning rate of 2e-5 for 1 epoch in mixed precision.

Cross-Lingual Alignment For the training of CLA adapters, we use 600K bilingual paraphrase data as explained in §3.2. Each adapter is trained with a sequence length of 128, batch size of 256 and learning rate of 2e-5 for 1 epoch in mixed precision. We use the parallel adapter (He et al., 2022) with default settings in `Adapters` (Poth et al., 2023) for CLA training.

C Datasets

We provide detailed information on the training and evaluation datasets. The datasets are used only for research purposes in this work.

C.1 Paraphrase Data

Table 7 provides an overview of the paraphrase datasets used for training. The XNLI dataset is licensed with cc-by-nc-4.0. For the sources of other datasets, please refer to the information page.[9]

C.2 STS/STR Evaluation Data

We use the test split of the datasets for zero-shot evaluation. In the following, we list the sources of STS/STR data for all individual languages and language pairs. Note that for symmetric pairs (e.g. en-de and de-en), the score in our experiments is the average of both directions.

STS17 The data for en, ar, es, en-ar, en-tr and es-en in our STS17 comes from the original STS17 (Cer et al., 2017). The data for de, fr, cs, de-en, en-fr, en-cs, cs-en, de-fr, fr-de, cs-de, de-cs, cs-fr and fr-cs is created by Hercig and Kral (2021). And the en-de, fr-en, nl-en and it-en data is translated by Reimers and Gurevych (2020). Through combining the data from Hercig and Kral (2021) and Reimers and Gurevych (2019), we get evaluation sets for nl-de, nl-fr, nl-cs, it-de, it-fr and it-cs. All data except

[9] https://huggingface.co/datasets/sentence-transformers/embedding-training-data

Dataset	Description	Size
MNLI/XNLI	Multi-Genre NLI data. We build 128K (Anchor, Entailment, Contradiction) triplets using the original data.	128K
Sentence Compression	Pairs (long_text, compressed_text) from news articles.	108K
Simple Wiki	Matched pairs (English_Wikipedia, Simple_English_Wikipedia).	102K
Altlex	Matched pairs (English_Wikipedia, Simple_English_Wikipedia).	113K
Quora Duplicate Questions	Duplicate question pairs from Quora. We use the "triple" subset.	102K

Table 7: Overview of paraphrase datasets. Except for XNLI, all of them are English datasets and are machine-translated into our target languages for training.

for ko are from the SNLI domain, containing 250 sentence pairs per language pair. The ko data is translated from the English STS benchmark (Cer et al., 2017) by Ham et al. (2020), containing 2850 pairs in various domains.

STSB Senel et al. (2024) translate the en data from the STS benchmark (Cer et al., 2017) into 5 Turkic languages: az, kk, ky, ug and uz. There are 800 test sentence pairs from various domains for each language. Since the other training data for Uyghur is written in the Arabic script, we transliterate the Cyrillic Uyghur data in the benchmark into the Arabic script using the Uyghur Multi-Script Converter.[10] The Turkic language data are combined with the dataset for ko (Ham et al., 2020) to get evaluation data for ko-en, ko-az, ko-ky, ko-ug and ko-uz.

SICK We use the SICK dataset in English (Marelli et al., 2014), Polish (Dadas et al., 2020), Dutch (Wijnholds and Moortgat, 2021) and Spanish (Araujo et al., 2022) and combine them to create cross-lingual evaluation data for en-pl, en-nl, en-es, pl-nl, pl-es and nl-es. The test set size is 4.91K for each language (pair).

STR24 We use the test data of the supervised track of STR24, including monolingual data for en (2600 pairs), am (342 pairs), ha (1206 pairs), rw (444 pairs), mr (298 pairs), te (297 pairs). We do not include Spanish because the public test set is not available, nor the Moroccan Arabic and Algerian Arabic because they are not supported by LaBSE. The data is curated primarily from news (Ousidhoum et al., 2024).

[10] https://github.com/neouyghur/Uyghur-Multi-Script-Converter

D Detailed Results

D.1 Semantic Textual Similarity / Relatedness

STS17 See detailed results of LaBSE-based models in Table 8. Results for en-cs, de-fr, cs-de, and cs-fr are calculated as the average of symmetric language pairs (e.g. de-fr is the average of de-fr and fr-de).

STSB See detailed results of LaBSE-based models in Table 9 and mE5-based models in Table 10. All cross-lingual results are the average of symmetric language pairs (e.g. az-kk is the average of az-kk and kk-az).

SICK See detailed results of LaBSE-based models in Table 11 and mE5-based models in Table 12. All cross-lingual results are the average of symmetric language pairs.

STR24 See detailed results of LaBSE-based models in Table 13.

D.2 Classification

SIB See detailed results of LaBSE-based models in Table 14 and mE5-based models in Table 15.

D.3 Bitext Mining

FLORES For LaBSE and mE5, we report the result of the best Full variant (Full_{c} for LaBSE in Table 16 and Full_{mc} for mE5 in Table 18) and our $\text{Mod}_{\text{mc-jt}}$ model (LaBSE-based in Table 17, mE5-based in Table 19) on each language pair.

Tatoeba See detailed results of LaBSE-based models in Table 20 and mE5-based models in Table 21. The results are the average of both directions (e.g. az the average of en-az and az-en).

	en	ar	cs	de	es	fr	ko	avg
LaBSE	79.4	69.1	79.1	79.3	80.8	77.9	71.3	76.7
Full_{en}	84.9	78.7	82.5	82.8	86.0	82.6	81.4	82.7
Full_{m}	85.0	77.5	83.8	83.9	86.7	82.5	80.9	82.9
Full_{c}	82.2	77.3	80.9	81.2	86.3	79.9	79.3	81.0
Full_{mc}	81.5	76.0	80.2	79.7	85.1	78.1	79.3	80.0
Mod_{en}	85.8	78.3	82.7	83.0	85.6	81.8	81.3	82.6
Mod_{m}	85.8	78.4	83.6	83.7	86.1	81.9	82.2	83.1
$\text{Mod}_{\text{mc-pp}}$	85.8	78.1	83.3	83.4	85.9	81.7	82.4	82.9
$\text{Mod}_{\text{mc-pl}}$	85.8	77.6	80.3	82.0	84.2	79.3	80.8	81.4
$\text{Mod}_{\text{mc-jt}}$	85.8	78.0	82.4	83.7	85.2	81.6	82.5	82.7

	en-ar	en-cs	en-de	en-es	en-fr	en-it	en-nl	en-tr	cs-de	cs-fr	de-fr	it-cs	it-de	it-fr	nl-cs	nl-de	nl-fr	avg
LaBSE	74.5	78.0	75.3	65.7	77.0	77.0	75.2	72.1	77.1	75.6	75.6	75.9	71.5	75.4	75.0	72.4	73.8	74.5
Full_{en}	79.6	80.9	81.2	75.1	81.6	81.4	80.6	75.6	78.8	77.8	78.0	78.1	76.9	79.7	78.7	77.6	77.7	78.8
Full_{m}	79.8	82.6	81.4	75.5	82.0	81.0	80.9	73.8	80.6	79.3	79.0	78.9	77.4	79.7	80.3	78.3	78.7	79.4
Full_{c}	78.4	80.6	79.2	74.3	79.8	79.8	79.3	74.9	78.8	77.2	76.8	77.3	75.9	78.2	77.9	76.5	77.2	77.8
Full_{mc}	79.6	79.4	77.8	75.0	78.0	77.3	77.6	76.5	77.4	76.4	75.4	76.3	73.9	76.6	76.5	75.0	75.1	76.7
Mod_{en}	79.5	81.9	82.4	76.0	81.8	82.7	82.3	77.1	80.5	78.2	79.7	80.0	79.6	80.5	79.6	79.9	79.8	80.1
Mod_{m}	80.6	82.4	82.4	77.0	81.8	83.3	82.6	77.5	81.2	79.0	80.0	80.5	80.4	80.6	80.0	80.4	79.8	80.6
$\text{Mod}_{\text{mc-pp}}$	81.5	82.3	82.2	76.8	82.0	83.6	82.6	77.5	81.2	79.0	80.0	81.4	80.4	81.3	80.5	80.2	79.9	80.7
$\text{Mod}_{\text{mc-pl}}$	80.1	81.1	81.2	76.7	81.2	82.3	81.6	76.5	78.6	77.4	77.9	78.8	78.6	79.1	77.1	78.1	77.9	79.1
$\text{Mod}_{\text{mc-jt}}$	81.4	81.8	82.3	77.5	81.9	83.3	82.1	77.3	80.6	78.8	80.0	80.2	80.3	80.8	78.8	79.5	79.0	80.3

Table 8: Results of LaBSE-based models on STS17.

	en	az	kk	ky	ug	uz	avg
LaBSE	72.5	70.8	78.0	70.6	69.6	69.6	71.9
Full_{en}	84.3	81.3	84.8	77.4	78.3	79.2	80.9
Full_{m}	84.2	80.1	84.4	76.7	77.5	79.5	80.4
Full_{c}	82.5	78.2	83.4	75.9	76.2	78.2	79.1
Full_{mc}	83.0	78.8	83.1	76.2	76.0	78.2	79.2
Mod_{en}	85.5	81.2	84.5	80.9	79.7	80.6	82.1
Mod_{m}	85.5	81.6	84.6	80.8	79.0	80.8	82.0
$\text{Mod}_{\text{mc-pp}}$	85.5	80.9	84.3	79.7	79.7	80.4	81.8
$\text{Mod}_{\text{mc-pl}}$	85.5	80.6	84.0	80.0	79.1	80.2	81.6
$\text{Mod}_{\text{mc-jt}}$	85.5	81.5	84.5	80.1	80.2	81.0	82.1

	en-az	en-kk	en-ko	en-ky	en-ug	en-uz	az-kk	az-ko	az-ky	az-ug	az-uz	kk-ko	kk-ky	kk-ug	kk-uz	ky-ko	ky-ug	ky-uz	ug-ko	ug-uz	uz-ko	avg
LaBSE	68.6	70.3	64.8	67.1	66.0	65.4	68.4	62.2	64.6	63.6	64.1	62.6	69.4	65.1	67.4	60.6	63.2	63.6	58.4	59.3	57.6	64.4
Full_{en}	75.8	76.5	73.2	73.6	71.1	72.4	75.2	69.6	72.5	69.5	71.6	70.4	76.0	71.6	73.5	67.6	69.6	70.8	66.2	67.6	66.4	71.5
Full_{m}	76.0	77.4	74.4	73.8	71.8	72.6	74.8	70.2	71.8	68.8	70.9	70.6	75.6	71.6	74.2	68.2	68.6	70.4	66.0	67.3	66.9	71.5
Full_{c}	76.2	77.6	73.8	74.9	72.5	75.1	74.6	70.0	72.2	68.4	71.7	70.8	75.4	71.4	74.6	69.2	69.5	71.9	66.6	68.2	68.8	72.1
Full_{mc}	76.8	78.2	74.8	75.1	73.6	75.5	75.2	71.0	72.8	69.8	72.7	71.6	75.5	71.8	74.6	69.3	69.4	72.0	68.0	69.3	69.2	72.7
Mod_{en}	78.1	78.8	75.4	76.8	74.7	77.8	77.4	72.0	75.6	71.6	76.5	72.4	79.1	73.6	78.3	71.2	71.6	76.7	68.4	72.6	71.6	74.8
Mod_{m}	78.6	79.4	76.1	77.1	74.7	78.0	78.3	73.3	76.3	72.6	76.9	73.4	79.4	74.0	79.4	71.9	71.6	76.9	69.3	72.4	72.4	75.3
$\text{Mod}_{\text{mc-pp}}$	79.0	79.5	77.0	77.2	75.6	78.6	78.0	74.2	75.8	73.6	77.4	74.6	79.0	75.4	79.1	72.8	73.0	76.6	71.3	74.3	73.9	76.0
$\text{Mod}_{\text{mc-pl}}$	79.1	80.0	76.6	77.7	75.8	78.8	78.0	73.8	76.0	73.8	77.4	74.8	79.0	75.8	79.0	72.7	73.2	76.6	71.2	74.2	73.6	76.1
$\text{Mod}_{\text{mc-jt}}$	79.2	80.2	77.0	77.6	76.1	79.0	78.6	74.4	76.2	74.3	77.5	75.1	79.2	75.9	79.4	72.8	73.5	76.8	71.7	74.8	74.1	76.4

Table 9: Results of LaBSE-based models on STSB.

	en	az	kk	ky	ug	uz	avg
mE5	85.2	72.7	75.6	67.5	63.0	71.2	72.5
Full_{en}	86.8	77.1	79.4	71.6	66.4	73.3	75.8
Full_{m}	86.6	79.5	83.4	75.9	74.4	77.7	79.6
Full_{c}	84.9	76.9	81.3	74.3	73.7	75.0	77.7
Full_{mc}	84.2	76.5	80.8	74.2	73.8	75.1	77.4
Mod_{en}	86.3	78.0	82.8	76.9	75.6	79.6	79.9
Mod_{m}	86.3	80.8	84.6	79.9	79.2	81.7	82.1
$\text{Mod}_{\text{mc-pp}}$	86.3	79.8	84.4	79.1	79.2	81.5	81.7
$\text{Mod}_{\text{mc-pl}}$	86.3	78.7	83.2	77.7	77.9	81.3	80.8
$\text{Mod}_{\text{mc-jt}}$	86.3	79.9	84.6	78.8	79.5	82.2	81.9

	en-az	en-kk	en-ko	en-ky	en-ug	en-uz	az-kk	az-ko	az-ky	az-ug	az-uz	kk-ko	kk-ky	kk-ug	kk-uz	ky-ko	ky-ug	ky-uz	ug-ko	ug-uz	uz-ko	avg
mE5	62.4	62.8	59.4	54.4	44.0	56.4	64.4	50.8	60.6	47.3	62.7	53.2	63.0	52.4	64.2	47.6	47.4	59.6	32.2	45.6	45.2	54.1
Full_{en}	63.2	62.6	63.6	56.6	43.2	56.9	65.3	56.3	60.6	45.4	62.0	57.0	65.5	50.3	63.8	52.4	47.2	58.4	39.6	43.2	49.6	55.4
Full_{m}	65.8	66.2	64.5	61.0	53.1	60.8	67.6	59.0	64.0	54.4	63.7	59.4	69.8	57.2	66.8	56.4	54.9	62.4	49.6	53.8	53.8	60.2
Full_{c}	72.4	73.1	69.7	69.0	66.6	69.5	71.4	64.6	67.9	62.6	67.6	64.8	71.2	65.9	71.1	62.2	61.8	66.2	59.4	62.7	61.2	66.7
Full_{mc}	71.9	72.4	69.8	68.7	65.4	70.2	71.1	64.9	67.8	61.9	68.2	64.6	70.9	65.2	71.0	63.0	62.4	67.2	59.6	62.2	62.0	66.7
Mod_{en}	69.6	70.9	67.9	69.1	63.9	71.6	71.4	61.4	68.8	61.2	69.8	62.2	72.8	64.2	72.4	60.9	63.1	69.4	55.3	62.1	62.4	66.2
Mod_{m}	72.9	73.4	68.5	72.4	69.2	74.9	74.2	64.0	72.2	66.9	73.2	64.4	75.5	69.0	75.6	63.8	68.1	73.7	60.4	67.9	64.8	69.8
$\text{Mod}_{\text{mc-pp}}$	75.2	76.2	73.0	74.8	72.4	77.2	75.6	69.2	73.4	70.6	74.8	70.2	76.8	73.1	77.5	69.3	71.5	75.8	67.6	72.4	70.7	73.2
$\text{Mod}_{\text{mc-pl}}$	75.0	76.0	72.7	74.6	71.6	77.4	74.8	69.2	72.8	69.9	74.8	70.3	75.9	71.8	76.9	68.8	70.7	75.2	67.4	71.8	71.0	72.8
$\text{Mod}_{\text{mc-jt}}$	75.8	76.8	73.8	75.2	72.9	77.8	75.8	69.9	73.6	71.4	75.4	71.2	77.2	73.4	78.0	69.9	71.8	76.2	68.5	72.9	71.8	73.8

Table 10: Results of mE5-based models on STSB.

	en	es	nl	pl	avg		
LaBSE	69.8	68.6	67.8	65.9	68.0		
Full_{en}	78.5	76.5	75.9	75.0	76.5		
Full_{m}	78.3	76.4	76.0	74.8	76.4		
Full_{c}	77.4	76.0	74.6	72.5	75.1		
Full_{mc}	77.1	75.4	74.7	73.2	75.1		
Mod_{en}	78.6	77.0	76.2	73.2	76.2		
Mod_{m}	78.6	76.7	76.4	74.4	76.5		
$\text{Mod}_{\text{mc-pp}}$	78.6	77.0	76.7	74.4	76.7		
$\text{Mod}_{\text{mc-pl}}$	78.6	76.3	75.7	73.5	76.0		
$\text{Mod}_{\text{mc-jt}}$	78.6	77.1	76.6	74.2	76.6		
	en-es	**en-nl**	**en-pl**	**es-nl**	**es-pl**	**nl-pl**	**avg**
LaBSE	65.2	65.6	65.2	63.7	61.0	61.9	63.8
Full_{en}	73.0	73.4	71.8	69.0	67.0	68.0	70.4
Full_{m}	73.4	74.0	71.9	69.8	67.8	68.5	70.9
Full_{c}	74.1	74.0	72.0	70.9	68.8	69.2	71.5
Full_{mc}	73.8	73.9	72.2	71.2	69.6	69.6	71.7
Mod_{en}	74.4	74.2	70.8	72.0	68.7	69.0	71.5
Mod_{m}	74.0	74.3	71.6	71.9	69.2	70.2	71.9
$\text{Mod}_{\text{mc-pp}}$	74.7	74.7	72.3	73.0	70.7	71.2	72.8
$\text{Mod}_{\text{mc-pl}}$	74.3	74.4	71.8	72.7	70.5	70.9	72.4
$\text{Mod}_{\text{mc-jt}}$	74.6	74.8	72.4	72.9	70.6	71.2	72.7

Table 11: Results of LaBSE-based models on SICK.

	en	es	nl	pl	avg		
mE5	77.9	75.3	72.5	71.2	74.2		
Full_{en}	79.0	76.4	74.0	72.0	75.4		
Full_{m}	79.3	76.3	74.6	71.9	75.5		
Full_{c}	77.7	74.9	73.2	69.9	73.9		
Full_{mc}	77.1	74.3	72.1	68.9	73.1		
Mod_{en}	78.5	76.3	75.4	73.1	75.8		
Mod_{m}	78.5	76.6	76.2	73.9	76.3		
$\text{Mod}_{\text{mc-pp}}$	78.5	76.6	76.5	73.9	76.4		
$\text{Mod}_{\text{mc-pl}}$	78.5	74.9	74.8	72.4	75.2		
$\text{Mod}_{\text{mc-jt}}$	78.5	76.7	76.4	73.8	76.4		
	en-es	**en-nl**	**en-pl**	**es-nl**	**es-pl**	**nl-pl**	**avg**
mE5	69.2	62.4	58.3	59.9	58.2	58.2	61.0
Full_{en}	69.4	65.5	60.2	62.5	58.8	57.0	62.2
Full_{m}	70.1	67.0	62.1	63.8	61.6	60.0	64.1
Full_{c}	72.6	69.9	66.4	67.9	65.2	64.3	67.7
Full_{mc}	71.9	69.4	65.6	67.2	64.2	63.3	66.9
Mod_{en}	69.2	70.2	65.4	66.9	63.4	64.9	66.7
Mod_{m}	69.9	71.4	67.9	68.6	65.7	67.6	68.5
$\text{Mod}_{\text{mc-pp}}$	71.4	72.5	69.6	70.9	68.6	69.8	70.5
$\text{Mod}_{\text{mc-pl}}$	70.8	71.5	68.4	70.1	67.9	69.0	69.6
$\text{Mod}_{\text{mc-jt}}$	71.7	72.5	69.8	71.1	68.9	70.2	70.7

Table 12: Results of mE5-based models on SICK.

	en	am	ha	mr	rw	te	avg
LaBSE	81.8	78.5	47.7	81.8	45.3	80.2	69.2
Full_{en}	80.6	79.9	63.8	86.1	57.2	84.7	75.4
Full_{m}	80.3	81.1	63.3	87.8	58.9	84.2	75.9
Full_{c}	80.1	81.1	63.0	86.5	57.7	83.2	75.3
Full_{mc}	80.5	80.4	62.8	87.3	58.9	82.7	75.4
Mod_{en}	82.3	82.7	67.3	86.3	67.5	86.3	78.7
Mod_{m}	82.3	83.1	67.8	87.3	63.2	86.4	78.4
$\text{Mod}_{\text{mc-pp}}$	82.3	83.6	66.5	87.0	59.2	86.1	77.4
$\text{Mod}_{\text{mc-pl}}$	82.3	83.1	66.1	85.3	61.8	84.7	77.2
$\text{Mod}_{\text{mc-jt}}$	82.3	84.0	67.1	87.1	62.6	85.4	78.1

Table 13: Results of LaBSE-based models on STR24.

	en	am	ar	az	cs	de	fr	ha	it	kk	rw	ky	ko	mr	nl	pl	ru	es	te	tr	ug	uz	zh	avg
monolingual fine-tuning																								
LaBSE	85.8	80.4	83.8	79.4	85.3	84.3	83.8	77.9	84.3	81.4	77.5	86.3	82.8	81.4	81.9	85.8	83.3	83.3	85.3	80.9	81.9	78.4	85.8	82.7
Full_{en}	86.8	77.0	84.3	83.3	86.3	87.3	88.7	75.5	87.8	83.3	80.4	86.3	82.4	83.8	87.3	87.3	85.8	85.8	87.3	85.8	79.9	77.9	84.8	84.1
Full_{m}	86.3	80.9	85.8	84.8	86.3	87.3	86.3	74.5	83.3	83.3	82.4	84.3	84.8	84.8	86.8	85.8	87.3	85.8	85.3	84.8	86.3	83.8	88.7	84.8
Full_{c}	86.3	81.4	87.8	84.8	84.8	85.8	88.2	77.5	84.3	82.8	79.9	85.8	87.3	85.3	86.3	87.8	87.8	85.8	87.8	85.8	85.3	82.4	86.3	85.1
Full_{mc}	89.2	81.4	86.3	88.2	86.3	85.8	87.8	79.4	87.8	82.8	83.3	85.8	84.3	85.3	89.2	88.2	89.2	86.8	87.3	87.3	89.7	80.9	85.8	86.0
Mod_{en}	87.8	84.8	84.3	86.3	87.8	87.3	87.3	79.4	83.8	85.8	83.3	86.8	85.8	83.8	83.3	83.8	84.8	84.8	83.8	86.8	83.3	83.3	84.3	84.9
Mod_{m}	87.8	80.9	85.8	86.8	87.8	88.2	88.2	81.9	86.3	85.8	82.4	86.8	84.3	84.3	84.8	85.3	86.3	84.3	84.8	88.2	83.8	86.8	86.3	85.6
$\text{Mod}_{\text{mc-pp}}$	87.8	84.3	86.8	88.2	86.8	88.2	85.3	82.4	88.2	85.8	82.4	87.8	84.8	85.8	84.8	86.3	86.8	87.3	84.8	86.3	84.3	85.8	86.3	86.0
$\text{Mod}_{\text{mc-pl}}$	87.8	81.9	87.8	87.3	88.2	88.2	87.8	82.8	86.3	84.8	84.3	86.3	84.3	86.3	86.3	84.3	85.8	84.8	87.3	87.8	85.3	82.8	84.3	85.8
$\text{Mod}_{\text{mc-jt}}$	87.8	82.4	85.8	87.3	88.2	88.7	87.3	81.9	85.8	88.2	81.9	85.3	83.8	84.3	85.3	85.8	86.3	85.8	86.3	88.2	86.8	83.8	85.8	85.8
cross-lingual transfer																								
LaBSE	85.8	81.4	83.8	81.9	86.8	84.3	87.8	80.9	85.8	82.4	79.4	84.8	81.9	80.9	85.8	85.3	82.8	86.8	85.3	81.9	80.9	80.9	85.8	83.6
Full_{en}	86.8	77.9	85.3	84.8	84.3	85.3	86.8	77.9	83.8	83.8	77.0	83.8	85.3	81.9	87.8	83.3	83.8	82.4	84.8	85.3	80.9	82.8	85.8	83.5
Full_{m}	86.3	76.5	85.3	87.3	84.8	85.3	86.3	76.5	84.3	84.8	78.9	85.8	84.3	81.4	85.8	82.8	86.3	84.8	84.3	86.8	83.3	81.4	84.3	83.8
Full_{c}	86.3	82.8	87.8	85.3	84.8	85.8	88.7	79.4	84.8	85.8	81.4	87.8	87.3	83.3	85.8	86.3	86.8	84.3	87.3	87.3	85.3	83.3	84.3	85.3
Full_{mc}	89.2	81.9	88.2	87.3	85.3	89.2	89.7	78.4	87.3	86.3	82.8	87.3	84.8	85.8	87.8	87.3	87.8	88.2	87.3	88.2	86.3	82.8	84.8	86.3
Mod_{en}	87.8	81.9	85.3	82.4	84.3	87.3	86.8	78.9	85.8	85.8	82.8	83.8	85.3	79.4	84.8	82.8	85.3	84.3	82.8	83.8	79.4	80.9	81.9	83.6
Mod_{m}	87.8	79.9	83.8	85.3	86.3	86.3	89.7	79.4	86.8	85.3	84.8	88.2	85.3	84.3	85.8	84.8	85.3	85.3	85.3	84.3	84.8	84.3	81.9	85.0
$\text{Mod}_{\text{mc-pp}}$	87.8	83.3	84.8	85.3	87.3	86.8	87.8	78.9	86.3	85.8	83.8	86.8	83.8	83.8	86.3	84.8	86.3	85.3	83.8	83.8	84.3	84.3	84.3	85.0
$\text{Mod}_{\text{mc-pl}}$	87.8	83.3	84.8	85.8	87.3	87.3	89.7	84.3	87.8	86.8	84.3	87.3	84.8	86.3	86.8	85.8	87.8	87.8	86.8	85.3	85.8	84.3	85.3	86.2
$\text{Mod}_{\text{mc-jt}}$	87.8	83.8	85.3	86.8	86.8	87.8	88.7	82.4	87.3	86.3	84.8	87.3	83.3	86.3	84.8	84.3	85.8	85.8	86.3	87.3	83.8	84.8	84.8	85.7

Table 14: Results of LaBSE-based models on SIB.

	en	az	kk	ky	ko	nl	pl	es	ug	uz	avg
monolingual fine-tuning											
mE5	83.3	73.5	76.0	73.5	77.5	77.5	77.5	70.1	59.8	71.6	74.0
Full_{en}	91.2	83.3	82.8	79.9	81.9	90.2	86.8	87.3	73.0	77.5	83.4
Full_{m}	89.7	85.3	84.3	85.8	82.8	88.2	87.8	85.8	81.4	83.8	85.5
Full_{c}	86.8	85.3	84.8	85.3	85.3	87.3	89.2	87.8	84.3	79.9	85.6
Full_{mc}	88.2	85.8	84.8	85.3	84.3	87.3	87.3	86.8	84.8	79.9	85.4
Mod_{en}	87.3	88.7	88.2	88.7	82.4	87.8	87.8	89.2	84.8	85.3	87.0
Mod_{m}	87.3	88.7	87.3	89.7	84.8	88.2	87.8	90.2	86.3	88.2	87.8
$\text{Mod}_{\text{mc-pp}}$	87.3	88.7	88.7	87.3	87.8	88.2	89.2	88.7	87.8	85.8	87.9
$\text{Mod}_{\text{mc-pl}}$	87.3	89.2	88.2	86.8	88.7	87.8	89.7	90.7	88.7	88.2	88.5
$\text{Mod}_{\text{mc-jt}}$	87.3	88.7	86.8	87.3	88.7	87.3	90.2	92.2	87.3	87.3	88.3
cross-lingual transfer											
mE5	83.3	72.1	71.1	70.6	75.5	77.5	78.9	78.9	58.8	68.6	73.5
Full_{en}	91.2	81.4	83.8	82.8	81.4	88.2	86.3	84.3	69.1	80.9	82.9
Full_{m}	89.7	86.3	85.8	82.8	83.8	87.3	88.2	85.8	77.5	85.3	85.2
Full_{c}	86.8	86.8	86.3	85.8	84.3	85.8	85.3	84.8	84.3	84.8	85.5
Full_{mc}	88.2	86.8	87.3	87.3	84.8	87.3	87.3	85.8	85.3	84.8	86.5
Mod_{en}	87.3	90.2	86.8	89.2	84.8	85.3	88.2	85.8	86.8	85.3	87.0
Mod_{m}	87.3	89.2	88.7	88.2	85.8	86.3	88.2	86.8	87.3	89.2	87.7
$\text{Mod}_{\text{mc-pp}}$	87.3	90.2	86.8	89.2	85.3	87.8	87.8	87.8	87.3	86.8	87.6
$\text{Mod}_{\text{mc-pl}}$	87.3	90.2	89.7	88.7	89.2	87.8	90.2	88.7	89.7	88.7	89.0
$\text{Mod}_{\text{mc-jt}}$	87.3	87.8	89.7	87.3	89.2	88.2	89.7	87.8	87.8	88.7	88.3

Table 15: Results of mE5-based models on SIB.

	am	ar	az	cs	de	en	fr	ha	it	kk	rw	ky	ko	mr	nl	pl	ru	es	te	tr	ug	uz	zh
am	-	0	0.2	0.1	0	0	0	0.79	0	0.2	0.69	0.2	0.1	0.1	0	0	0	0.1	0	0.1	0.1	0.1	0.1
ar	0	-	0.2	0	0	0	0	0.79	0	0.2	0.49	0.1	0	0.1	0	0	0	0.1	0	0.1	0.2	0.1	0
az	0.2	0.3	-	0.1	0.1	0.2	0.1	0.99	0.1	0.4	1.38	0.3	0.2	0.49	0.1	0.2	0.3	0.2	0.2	0.2	0.3	0.3	0.49
cs	0.1	0	0.2	-	0	0	0	0.89	0	0.2	0.49	0.1	0	0.1	0	0	0	0.1	0	0.1	0.2	0.1	0
de	0	0	0.3	0	-	0	0.1	0.89	0.1	0.2	0.79	0.2	0.1	0.1	0	0	0	0.1	0.1	0.1	0.1	0.1	0.1
en	0	0	0.2	0	0	-	0	0.4	0	0.2	0.49	0	0	0	0	0	0	0.1	0	0.1	0.1	0.1	0
fr	0	0	0.2	0	0	0	-	0.49	0	0.2	0.49	0.1	0.1	0.1	0	0	0	0.2	0.1	0	0.1	0.1	0.1
ha	0.79	0.4	0.99	0.59	0.4	0.3	0.4	-	0.4	0.79	1.48	0.69	0.79	0.4	0.4	0.59	0.49	0.49	0.49	0.49	0.69	0.69	0.49
it	0.1	0	0.3	0	0	0	0	0.49	-	0.2	0.69	0	0.1	0.1	0.1	0	0	0.1	0	0.1	0.1	0.1	0
kk	0.2	0.2	0.4	0.2	0.2	0.2	0.2	0.99	0.2	-	1.38	0.3	0.3	0.3	0.3	0.2	0.2	0.3	0.2	0.2	0.4	0.3	0.3
rw	0.69	0.49	0.79	0.3	0.49	0.3	0.4	1.48	0.3	1.09	-	0.49	0.69	0.79	0.49	0.69	0.69	0.69	0.89	0.59	1.09	0.59	0.49
ky	0.1	0.1	0.2	0.1	0.1	0	0	0.69	0	0.2	0.79	-	0.4	0.1	0.1	0	0	0.1	0	0.1	0.2	0.1	0.1
ko	0.3	0	0.3	0	0	0	0	0.79	0	0.3	0.99	0.3	-	0.1	0	0	0.2	0.1	0.1	0.1	0.2	0.1	0
mr	0	0	0.2	0.1	0	0	0	0.59	0	0.3	0.49	0.1	0.1	-	0	0	0	0.1	0	0.1	0.1	0.1	0.1
nl	0.1	0	0.4	0.1	0	0	0	0.69	0	0.3	0.69	0.1	0.2	0.1	-	0	0	0.2	0	0.1	0.3	0.1	0.1
pl	0.2	0	0.4	0	0	0	0	0.59	0	0.3	0.59	0	0	0.2	0.1	-	0	0.1	0.1	0	0.2	0.1	0
ru	0	0	0.3	0	0	0	0	0.59	0.1	0.2	0.59	0	0.2	0.1	0	0	-	0.1	0	0	0.2	0.1	0
es	0.1	0.1	0.2	0.1	0.1	0.1	0.1	0.69	0.1	0.3	0.69	0.1	0.1	0.2	0.1	0.1	0.2	-	0.1	0.1	0.2	0.2	0.1
te	0.1	0	0.2	0	0	0	0.1	0.4	0	0.3	0.79	0	0.1	0	0	0.2	0	0.2	-	0.1	0.1	0.1	0
tr	0	0	0.1	0	0	0	0	0.99	0	0.2	0.59	0	0	0	0.1	0	0	0.1	0	-	0.2	0.1	0
ug	0.1	0.1	0.4	0.1	0.1	0.1	0.1	0.89	0.1	0.4	0.99	0.4	0.2	0.1	0.2	0.1	0.2	0.3	0.2	0.3	-	0.2	0.1
uz	0.1	0.1	0.3	0.1	0.1	0.1	0.1	0.79	0.1	0.4	0.89	0.1	0.1	0.2	0.1	0.1	0.1	0.2	0.2	0.3	0.2	-	0.2
zh	0.2	0	0.3	0	0	0	0	0.89	0	0.4	0.79	0.2	0.1	0.3	0	0	0	0.1	0	0.2	0.3	0.1	-

Table 16: Results of LaBSE-based Full_{c} baseline on FLORES.

	am	ar	az	cs	de	en	fr	ha	it	kk	rw	ky	ko	mr	nl	pl	ru	es	te	tr	ug	uz	zh
am	-	0	0.3	0	0	0	0	0.4	0	0.2	0.4	0.1	0	0	0	0	0	0.1	0	0.1	0.1	0.1	0
ar	0.1	-	0.4	0.1	0.1	0	0	0.49	0.1	0.2	0.49	0.1	0.1	0.2	0.1	0.1	0.1	0.1	0	0.2	0.2	0.2	0
az	0.4	0.3	-	0.2	0.2	0.2	0.2	0.79	0.2	0.4	0.69	0.2	0.2	0.3	0.2	0.3	0.2	0.3	0.2	0.2	0.3	0.3	0.2
cs	0	0	0.2	-	0	0	0	0.4	0	0.2	0.49	0	0	0.1	0	0	0	0.1	0	0	0.1	0.1	0
de	0	0	0.2	0	-	0	0	0.49	0	0.2	0.3	0	0.1	0.1	0	0	0	0.1	0.1	0	0.1	0.1	0
en	0	0	0.2	0	0	-	0	0.4	0	0.2	0.4	0	0	0	0	0	0	0.1	0	0	0.1	0.1	0
fr	0	0	0.2	0	0	0	-	0.4	0	0.2	0.4	0	0	0	0	0	0	0.1	0	0	0.1	0.1	0
ha	0.3	0.49	0.89	0.4	0.3	0.4	0.4	-	0.4	0.59	0.59	0.49	0.59	0.49	0.59	0.49	0.4	0.49	0.4	0.59	0.49	0.49	0.4
it	0	0	0.2	0	0	0	0	0.3	-	0.2	0.3	0	0	0.1	0.1	0	0	0.1	0	0	0.1	0.1	0
kk	0.4	0.2	0.4	0.2	0.2	0.2	0.2	0.69	0.2	-	0.59	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.2	0.3	0.3	0.3	0.2
rw	0.49	0.4	0.49	0.2	0.2	0.2	0.2	0.49	0.2	0.49	-	0.3	0.2	0.2	0.3	0.2	0.2	0.3	0.4	0.2	0.3	0.3	0.2
ky	0	0.1	0.2	0	0	0	0.1	0.69	0	0.3	0.4	-	0.1	0	0	0	0	0.1	0	0	0.1	0.2	0.1
ko	0.1	0	0.2	0.1	0	0	0	0.69	0	0.2	0.79	0.2	-	0.1	0	0	0	0.1	0	0.1	0.2	0.1	0
mr	0	0.1	0.3	0.1	0.1	0	0.1	0.59	0.1	0.2	0.3	0.1	0.1	-	0.1	0.1	0	0.1	0	0.1	0.1	0.1	0
nl	0	0.1	0.2	0	0	0	0	0.59	0	0.2	0.59	0	0	0.1	-	0.1	0	0.1	0	0	0.1	0.1	0
pl	0	0.1	0.4	0	0	0	0.1	0.59	0	0.2	0.4	0.1	0	0.1	0	-	0	0.2	0.1	0	0.1	0.1	0
ru	0	0.1	0.2	0	0	0	0	0.4	0	0.2	0.4	0	0.1	0.1	0	0	-	0.1	0.1	0	0.1	0.1	0
es	0.1	0.1	0.3	0.1	0.1	0.1	0.1	0.59	0.1	0.3	0.3	0.1	0.1	0.1	0.1	0.2	0.1	-	0.1	0.1	0.2	0.2	0.1
te	0	0	0.3	0	0	0	0.1	0.59	0	0.3	0.49	0.2	0	0.1	0	0.1	0	0.2	-	0.1	0.1	0.1	0
tr	0	0.1	0.2	0	0	0	0	0.69	0	0.2	0.3	0	0	0	0.1	0	0	0.1	0	-	0.1	0.1	0
ug	0.1	0.2	0.4	0.2	0.1	0.1	0.1	0.59	0.1	0.4	0.4	0.1	0.2	0.1	0.1	0.1	0.1	0.2	0.1	0.2	-	0.2	0.1
uz	0.1	0.2	0.3	0.1	0.1	0.1	0.1	0.79	0.1	0.3	0.3	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.2	0.2	-	0.1
zh	0.3	0	0.3	0	0	0	0	0.49	0	0.3	0.3	0.1	0	0.2	0	0	0	0.1	0	0.1	0.2	0.1	-

Table 17: Results of LaBSE-based $\text{Mod}_{\text{mc-jt}}$ on FLORES.

	az	en	kk	ky	ko	nl	pl	es	ug	uz
az	-	0.1	0.49	0.2	0.4	0.2	0.3	0.2	0.59	0.4
en	0.2	-	0.2	0	0.1	0	0	0.1	0.1	0.1
kk	0.4	0.2	-	0.3	0.4	0.2	0.3	0.4	0.49	0.3
ky	0.59	0	0.3	-	0.2	0.59	0.2	0.1	0.69	0.3
ko	0.59	0	0.49	0.49	-	0.1	0.3	0.1	0.79	0.4
nl	0.3	0	0.2	0.3	0.1	-	0	0.1	0.3	0.1
pl	0.4	0	0.4	0.1	0.2	0	-	0.1	0.4	0.1
es	0.3	0.1	0.49	0.1	0.2	0.1	0.1	-	0.2	0.2
ug	0.59	0.1	0.49	0.59	0.59	0.2	0.1	0.2	-	0.4
uz	0.4	0.1	0.4	0.3	0.2	0.1	0.1	0.3	0.49	-

Table 18: Results of mE5-based Full_{mc} baseline on FLORES.

	az	en	kk	ky	ko	nl	pl	es	ug	uz
az	-	0.1	0.59	0.4	0.3	0.3	0.4	0.3	0.4	0.4
en	0.3	-	0.2	0.1	0.1	0	0	0.1	0.1	0.1
kk	0.49	0.2	-	0.2	0.3	0.2	0.3	0.4	0.4	0.3
ky	0.3	0.1	0.2	-	0.1	0.2	0	0.1	0.1	0.1
ko	0.4	0	0.3	0.3	-	0	0	0.1	0.1	0.1
nl	0.3	0	0.2	0.1	0.1	-	0	0.1	0.1	0.1
pl	0.4	0	0.3	0.2	0.1	0	-	0.1	0.2	0.1
es	0.3	0.1	0.4	0.1	0.1	0.1	0.1	-	0.2	0.2
ug	0.59	0.1	0.4	0.2	0.1	0.2	0.2	0.2	-	0.2
uz	0.4	0.2	0.3	0.1	0.1	0.1	0.1	0.2	0.2	-

Table 19: Results of mE5-based $\text{Mod}_{\text{mc-jt}}$ on FLORES.

	az	kk	ug	uz	am	te	mr	cs	fr	de	ar	es	it	tr	pl	nl	zh	ru	ko	avg
LaBSE	3.10	7.74	5.35	10.86	4.76	0.85	4.45	2.05	3.55	0.40	7.60	1.30	4.60	1.00	1.30	1.90	3.10	4.15	5.40	3.87
Full_{en}	4.15	9.65	7.05	13.67	5.95	1.07	5.10	2.30	3.75	0.35	9.95	1.45	4.95	1.80	1.60	2.50	3.40	4.70	6.20	4.72
Full_{m}	4.10	8.26	5.85	10.40	6.55	1.71	4.45	2.45	4.00	0.30	9.95	1.65	4.65	1.95	1.60	2.60	2.95	4.60	6.15	4.43
Full_{c}	3.50	7.30	5.60	9.58	4.76	1.07	4.50	2.35	3.70	0.30	8.35	1.45	4.70	1.55	1.30	2.80	2.90	4.45	5.85	4.00
Full_{mc}	3.40	8.26	5.65	9.70	5.95	1.07	4.50	2.60	3.45	0.45	8.80	1.75	4.50	1.90	1.50	2.65	2.65	4.80	5.60	4.17
Mod_{en}	2.40	6.87	4.40	6.54	4.76	1.07	5.85	2.50	3.10	0.45	7.10	1.35	4.55	1.85	2.00	2.15	3.15	4.65	5.15	3.68
Mod_{m}	2.30	7.13	4.40	6.66	5.06	1.28	5.30	2.30	3.35	0.55	6.35	1.25	4.30	1.90	1.60	2.10	2.95	4.95	5.30	3.63
$\text{Mod}_{\text{mc-pp}}$	2.10	6.70	4.40	6.31	4.76	1.71	4.20	2.20	3.20	0.35	6.40	1.30	4.65	1.40	1.60	2.05	2.95	4.55	5.50	3.49
$\text{Mod}_{\text{mc-pl}}$	2.20	7.04	4.75	6.19	5.36	1.92	4.05	2.35	3.25	0.50	7.10	1.35	4.35	1.60	1.60	2.05	3.25	4.85	5.35	3.64
$\text{Mod}_{\text{mc-jt}}$	2.00	6.96	4.50	5.84	5.36	1.28	4.20	2.30	3.35	0.45	6.65	1.30	4.40	1.65	1.65	2.15	3.15	4.65	5.65	3.55

Table 20: Results of LaBSE-based models on Tatoeba.

	az	kk	ug	uz	ko	es	pl	nl	avg
mE5	7.75	13.74	18.45	20.09	10.25	1.95	2.70	4.15	9.89
Full_{en}	7.40	13.48	17.15	22.66	10.30	2.00	3.35	3.60	9.99
Full_{m}	6.65	12.87	12.85	13.08	9.00	2.05	3.05	3.35	7.86
Full_{c}	4.95	11.04	8.35	12.03	7.60	1.60	2.55	2.85	6.37
Full_{mc}	4.90	10.87	8.90	10.75	8.15	1.85	2.35	2.90	6.33
Mod_{en}	4.10	10.00	7.50	9.58	8.20	1.95	2.50	2.65	5.81
Mod_{m}	3.50	10.35	6.05	7.94	7.50	2.00	2.35	2.35	5.26
$\text{Mod}_{\text{mc-pp}}$	3.30	10.09	6.30	7.59	7.40	2.20	2.25	2.30	5.18
$\text{Mod}_{\text{mc-pl}}$	3.65	11.04	7.10	7.94	7.65	2.35	2.65	2.50	5.61
$\text{Mod}_{\text{mc-jt}}$	3.25	9.57	5.95	7.71	7.10	2.00	2.35	2.20	5.02

Table 21: Results of mE5-based models on Tatoeba.